

ROHIT SHIDID

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PROFESSIONAL SUMMARY

Software engineer and NYU M.S. Computer Engineering candidate with 1+ year of production experience at Citi. Builds tested ML pipelines and AI systems using Python, PySpark, RAG pipelines, and embedded/cloud infrastructure, with experience spanning automated evaluation, CI/CD, and full-stack deployment.

TECHNICAL SKILLS

Languages: Python, C++, Go

Frameworks and Data: FastAPI, PySpark 4.0, Apache Kafka, Redpanda, Parquet, Hive, pandas, NumPy, PyArrow, PostgreSQL, MinIO, ETL design

AI/ML: PyTorch, scikit-learn, XGBoost, MLflow, Ray Tune, ASHA early-stopping, distributed hyperparameter search, feature stores, model registries, RAG pipelines, hallucination detection, NLI cross-encoders, prompt engineering, fine-tuning, query rewriting, HyDE, FAISS, sentence-transformers, DeBERTa-v3, LangChain, LangGraph, MCP protocol, Claude/OpenAI/Groq APIs

Systems and Embedded: ARM Cortex-M4, CMSIS-DSP, BLE, STM32, digital signal processing, OpenCV

Infrastructure and Quality: Docker, Kubernetes, Chameleon Cloud, AWS (S3, Lambda, SageMaker), Git, GitHub Actions, CI/CD, Redis, Streamlit, unit and integration testing

PROFESSIONAL EXPERIENCE

New York University | Graduate Teaching Assistant

New York, NY | 09/2025 – Present

- Cut assignment turnaround time 40% by coordinating grading workflows across Operating Systems, AI Governance, Cyber Risk Resilience, and Information Systems Security Engineering alongside a co-TA.
- Raised student comprehension scores 30% (midterm performance) by partnering with the lead professor to identify and close recurring concept gaps across four graduate courses.

Citi | Software Engineer

Pune, India | 07/2024 – 08/2025

- Improved trader execution latency 40% by building SQL analytics pipelines and automated ETL data ingestion workflows.
- Cut manual intervention in stakeholder reporting 90% by automating operational data workflows, reducing processing time from hours to minutes.
- Delivered zero-downtime production releases by deploying clean, tested pipeline code through Git-based CI/CD workflows.

Elespa HEV | AI Engineer

Pune, India | 07/2023 – 02/2024

- Flagged component failure risk early by building a predictive time-series model over multi-channel vehicle sensor data that learned normal operating ranges and surfaced out-of-range readings for engineering review.
- Maintained a two-week sprint cadence with zero missed commitments across 8 cycles by iterating the anomaly-detection model and its sensor-processing pipeline with engineering stakeholders to validate flagged events against real vehicle behavior.

FinIQ Consulting | Software Developer

Pune, India | 05/2023 – 07/2023

- Cut downstream reporting errors ~50% by writing and deploying corrective SQL scripts that resolved root-cause schema mismatches in the data transfer pipeline.
- Doubled ticket closure rate (Jira cycle time) by building a root-cause analysis tool over daily data ingestion logs that eliminated repeat failure categories.

PROJECTS

Self-Healing RAG System | *Demo*

FAISS, Groq (Llama-3.3-70B), sentence-transformers, DeBERTa-v3, Streamlit

- Raised fully-grounded answer rate 12 points (69% to 81%) and cut hallucinated or ungrounded responses 40%, measured on the answerable subset of a 20-question evaluation set versus a standard RAG baseline, by building a retrieval-augmented generation pipeline with automated failure detection across both retrieval and generation stages and 4 healing strategies — query rewriting, HyDE, claim-targeted re-retrieval, and context expansion — that activate selectively based on which failure mode is detected.
- Built per-claim hallucination detection running entirely at zero API cost, verified by sentence-window premise matching across all 20 evaluation questions, by engineering a cross-encoder NLI pipeline using DeBERTa-v3 for claim verification and a separate query–passage relevance cross-encoder for retrieval-failure detection — with 8 unit tests covering every failure-detection and healing-strategy route, and a real-time healing log UI deployed as a permanently-hosted Streamlit demo on Hugging Face Spaces.

Research Copilot — Agentic RAG over Academic Literature

LanceDB, BM25, BGE embeddings, Claude API, Groq API

- Enforced citation-grounded literature synthesis over arXiv and Semantic Scholar papers with hard citation validation that flags any [n] marker not mapped to a retrieved source, by building a hybrid retrieval pipeline that fuses BM25 lexical search with BGE dense embeddings through Reciprocal Rank Fusion and reranks results with a bge-reranker-v2-m3 cross-encoder.
- Built a provider-agnostic architecture that routes across Claude, GPT-4o, and Llama-3.3-70B (via Groq), exposed through independent /ask, /verify, and /followup endpoints with per-stage ablation toggles for isolating each retrieval component's contribution, by designing a FastAPI backend with a section-aware PDF chunker that preserves page and section provenance for every ingested paper.

PromptLint — LLM Router and Prompt Linter | *Demo*

FastAPI, async SQLAlchemy, PostgreSQL, Fernet encryption

- Cut LLM spend 89% on a 480-prompt benchmark versus always routing to the most expensive model, by building a cost-aware routing API that classifies prompt difficulty and task type in ~200ms and dispatches across Claude, GPT, Gemini, and any OpenAI-compatible endpoint with automatic provider fallback.
- Delivered prompt-quality scoring before a single paid-model call, returning a 0–100 score, a predicted first-try success probability, and 11 rule-based checks with ranked remediation suggestions, by engineering a FastAPI and async SQLAlchemy backend with hashed API-key auth, Fernet-encrypted provider keys, and per-key rate limiting, validated by 181 automated tests in GitHub Actions CI and deployed on Render with a zero-cost Neon Postgres backend.

GDELT News Intelligence Pipeline | *Demo*

PyTorch, Plotly Dash

- Reached 92% F1 on event conflict classification (held-out test split) by training and tracking 8 models across classification, regression, anomaly detection, and forecasting tasks over 250K+ GDELT 2.0 news events per run ingested via parallel download and PySpark ETL into a Hive-style Parquet warehouse.
- Reached an MAE of 0.73 on 0-100 event severity prediction (held-out evaluation set) by training a PyTorch feed-forward network on engineered geopolitical and media-coverage features extracted from the partitioned Parquet store, with per-run MLOps metric logging across all model types.

Bandageboard — Full-Stack Healthcare Billing Pipeline | *Demo*

Next.js 14, Claude API, Vercel Postgres, Drizzle ORM

- Automated Medicare Part B wound-care billing triage for a hackathon challenge (300 synthetic patients, no real PHI), routing end to end from EHR-style ingestion to biller-facing decision output, by building a full-stack Next.js 14 application with a regex-first structured extraction pipeline and an Anthropic-powered fallback for unstructured clinical narratives — with schema-validated LLM outputs gating downstream eligibility computation, backed by Vercel Postgres via Drizzle ORM.
- Shipped a biller-facing React dashboard supporting full CRUD operations on structured billing records, demonstrating end-to-end ownership of an AI-augmented feature from ingestion pipeline through deployed UI, for the ABI Frameworks hackathon.

GemSpot — ML Training Subsystem

XGBoost, MLflow, Docker, Chameleon Cloud

- Tracked automated XGBoost model training across three MLflow-logged candidates — a majority-class baseline, default-parameter XGBoost, and a tuned XGBoost with class-imbalance handling — by containerizing the training pipeline with Docker and deploying it to a Chameleon Cloud VM.
- Reached 0.835–0.836 ROC-AUC on Google Maps review classification across the two XGBoost candidates by training on 337,698 real review records with a fixed-hyperparameter configuration defined in a single training config file.

OracleLoop — Autonomous, Self-Resolving Prediction Markets

Node.js, TypeScript, Gemini API, multi-agent orchestration

- Automated verified prediction-market question generation across live sports coverage, gated by a second, independent Gemini pass that rejects any candidate question found unsolvable before it ever reaches a user, by building a multi-agent pipeline where a research agent searches the web for live coverage of any sport or match, drafts a binary question from the fetched articles, and routes it through a dedicated verification agent before publish.
- Enabled prediction markets to close and settle with no manual input by generating a probability-weighted prediction for each published question, rendering it as a selectable two-option card with a live countdown timer, and routing expired cards to a dedicated resolution agent that re-searches the web and fetches the real-world result the moment the timer expires.

OmniRoute — LLM Gateway and Chat | Demo

Multi-Provider Routing, OpenAI-Compatible API, Self-Hosted Gateway

- Routed requests across 77+ LLMs from different providers through one OpenAI-compatible endpoint, so switching models never meant rewriting application code, by building a self-hosted gateway that normalizes each provider’s API into a single request format.
- Proved the routing worked end to end with a hosted chat client built on top of the gateway, switching between providers mid-conversation with live token streaming and no changes to the client code.

bak — Published Go CLI for Per-File Version Control (Homebrew)

Go, zstd, Homebrew, CLI tooling

- Delivered isolated per-file version control without full Git repositories or cluttered .bak fallback files, holding zero workspace clutter across all tracked files, by engineering a high-performance Go CLI with a centralized zstd-compressed storage engine that stores hundreds of file snapshots with minimal disk overhead.
- Enabled fast historical comparison and lifecycle management through interactive restore, –keep N pruning, and automated garbage collection via bak gc, by building a custom diffing engine producing unified and compact hunk-style diffs directly in the terminal, and authoring a custom Homebrew formula that fully automates the build and deployment pipeline for macOS users.

portmap — Published Go CLI (Homebrew)

Go, Homebrew, networking, CLI tooling

- Unified port visibility across local processes and containers, replacing 5 separate diagnostic commands (lsof, ss, netstat, docker ps, /etc/services), by building and publishing portmap — a lightweight Go CLI available through Homebrew that surfaces listening ports, owning processes, Docker container mappings, and known service labels in a single clean table or JSON output.

Open-Source Contribution — Odysseus

JavaScript, pytest, 85k★ self-hosted AI workspace

- Diagnosed a silent /todo slash-command failure and submitted a fix as a pull request with new pytest API-level regression coverage (closed by the maintainer without merging), by tracing a note_type payload mismatch in the JavaScript frontend (slashCommands.js) and adding structured list-item support and note_type filtering to the payload sent to the notes API.
- Prevented regression with a new pytest test file validating the corrected /todo payload shape through the notes API, by adding tests alongside the fix that catch the same class of frontend mismatch across the notes API surface.

EEG Independent Component Classification

scikit-learn, ensemble methods, signal processing

- Improved brain signal classification accuracy for cognitive research, per published evaluation metrics, by implementing ensemble-based ML techniques for EEG independent component analysis. Published: doi.org/10.5281/zenodo.13909560

Parkinson’s Motor Symptom Classifier — Embedded Signal Processing

ARM Cortex-M4, CMSIS-DSP, LSM6DSL IMU, BLE, STM32, I2C, DSP

- Delivered four-class real-time motor symptom classification, validated by correct state labeling across repeated bench-test trials, by reading a 6-axis IMU over I2C at 52 Hz, applying Hamming windowing and a 256-point real FFT via ARM CMSIS-DSP, and thresholding dominant frequencies into resting tremor (3-5 Hz), dyskinesia (5-7 Hz), freeze of gait, and normal walking (1-2 Hz) states — all on-device with no cloud dependency.
- Enabled live remote monitoring over BLE, verified by continuous streaming to a connected client during bench testing, by implementing a custom GATT service on the STM32 with automatic gyroscope calibration and moving-average filtering to suppress drift and noise before transmission.

Murmur — On-Device Dictation | Demo

Swift, macOS, On-Device ML, Speech-to-Text

- Kept every word of a dictation session on-device, with no audio or text ever leaving the machine, by building a push-to-talk macOS tool that runs speech-to-text locally from a single key hold to a finished sentence at the cursor.
- Smoothed out transcription errors by routing the raw speech-to-text output through an LLM pass that autocorrects and reframes sentences when needed, so the cleaned-up version is what actually lands at the cursor.

Synapse — Two-Answer Fact Checker | Demo

OpenRouter API, Web Search Integration, Node.js

- Settled disagreements between two competing answers to the same question using live web search for verification, rather than trusting either model’s own memory, by building a fact-checking pipeline on OpenRouter’s free-tier model routing.
- Left the verification layer dependency-light by handing all model routing and provider selection to OpenRouter, so the project’s own code covers only prompt construction, the web-search calls, and the final adjudication step.

Real-Time Cricket Ball Tracker | Demo

OpenCV, NumPy

- Delivered real-time cricket ball detection and trajectory tracking, holding stable frame-by-frame tracking across varied lighting and background conditions, by engineering a computer vision pipeline using tuned HSV color masking and contour analysis to isolate the ball and identify the exact bounce point.
- Automated delivery length classification into five categories — Yorker, Full, Good, Back of Length, Short — with correct pitch zone mapping across simulated delivery sequences, by implementing a 4-point homography calibration that transforms 2D image bounce coordinates into real-world pitch dimensions.

ADDITIONAL PROJECTS

Layoff Predictor, Disk Manager, AirControl, Hexagon, Splitplus, Mutation Simulator

EDUCATION

New York University, New York, NY Master of Science, Computer Engineering	Expected 05/2027
College of Engineering Pune, Pune, India Bachelor of Technology, Computer Engineering	05/2024